

My Project

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1 Hierarchical Index	1
1.1 Class Hierarchy	1
2 Class Index	3
2.1 Class List	3
3 File Index	5
3.1 File List	5
4 Class Documentation	7
4.1 Human Class Reference	7
4.1.1 Constructor & Destructor Documentation	8
4.1.1.1 Human() [1/4]	8
4.1.1.2 Human() [2/4]	8
4.1.1.3 Human() [3/4]	8
4.1.1.4 Human() [4/4]	8
4.1.1.5 ~Human()	8
4.1.2 Member Function Documentation	9
4.1.2.1 name()	9
4.1.2.2 operator=() [1/2]	9
4.1.2.3 operator=() [2/2]	9
4.1.2.4 setName()	9
4.1.2.5 setSurname()	9
4.1.2.6 surname()	9
4.1.3 Member Data Documentation	9
4.1.3.1 name_	9
4.1.3.2 surname_	9
4.2 Student Class Reference	10
4.2.1 Constructor & Destructor Documentation	11
4.2.1.1 Student() [1/6]	11
4.2.1.2 Student() [2/6]	11
4.2.1.3 Student() [3/6]	12
4.2.1.4 Student() [4/6]	12
4.2.1.5 Student() [5/6]	12
4.2.1.6 Student() [6/6]	12
4.2.1.7 ~Student()	12
4.2.2 Member Function Documentation	12
4.2.2.1 examGrade()	12
4.2.2.2 finalGradeAverage()	12
4.2.2.3 finalGradeMedian()	12
4.2.2.4 homeworkGrades()	12
4.2.2.5 name()	13
4.2.2.6 operator=() [1/2]	13

4.2.2.7 operator=() [2/2]	13
4.2.2.8 readStudent()	13
4.2.2.9 setExamGrade()	13
4.2.2.10 setHomeworkGrades()	13
4.2.2.11 setName()	13
4.2.2.12 setSurname()	13
4.2.2.13 surname()	13
4.2.3 Friends And Related Symbol Documentation	14
4.2.3.1 operator<<	14
4.2.3.2 operator>>	14
5 File Documentation	15
5.1 include/create-student.h File Reference	15
5.1.1 Function Documentation	16
5.1.1.1 createStudentFullyRandom()	16
5.1.1.2 createStudentManual()	16
5.1.1.3 createStudentRandomGrades()	16
5.1.1.4 createStudentsFromFile()	16
5.1.1.5 randomName()	16
5.1.1.6 randomSurname()	16
5.2 create-student.h	16
5.3 include/generate-student-list-test.h File Reference	17
5.3.1 Function Documentation	17
5.3.1.1 handleGenerateStudentListTest()	17
5.4 generate-student-list-test.h	17
5.5 include/generate-student-list.h File Reference	18
5.5.1 Function Documentation	18
5.5.1.1 generateStudentListFile()	18
5.5.1.2 handleGenerateStudentListFile()	18
5.6 generate-student-list.h	19
5.7 include/grading-utils.h File Reference	19
5.7.1 Function Documentation	19
5.7.1.1 calculateFinalGradeAverage()	19
5.7.1.2 calculateFinalGradeMedian()	20
5.7.1.3 randomGrade()	20
5.8 grading-utils.h	20
5.9 include/human.h File Reference	20
5.10 human.h	21
5.11 include/input-utils.h File Reference	21
5.11.1 Function Documentation	22
5.11.1.1 readIntInRange()	22
5.11.1.2 readSingleIntToken()	22

5.11.1.3 readSingleStringToken()	22
5.11.1.4 readYesOrNo()	22
5.12 input-utils.h	22
5.13 include/output-nuskriausti-and-protingi-test.h File Reference	23
5.13.1 Function Documentation	23
5.13.1.1 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles()	23
5.14 output-nuskriausti-and-protingi-test.h	23
5.15 include/output-results.h File Reference	23
5.15.1 Function Documentation	24
5.15.1.1 chooseOutputAndPrint()	24
5.15.1.2 outputResults()	24
5.16 output-results.h	25
5.17 include/sort-nuskriausti-and-protingi.h File Reference	25
5.17.1 Function Documentation	26
5.17.1.1 handleSortNuskriaustiAndProtingiFromFile()	26
5.17.1.2 sortNuskriaustiAndProtingiFirstStrategy()	26
5.17.1.3 sortNuskriaustiAndProtingiSecondStrategy()	26
5.17.1.4 sortNuskriaustiAndProtingiThirdStrategy()	26
5.18 sort-nuskriausti-and-protingi.h	26
5.19 include/sort-students.h File Reference	27
5.19.1 Function Documentation	27
5.19.1.1 runSortStudentsChoicePrompt()	27
5.19.1.2 sortStudents()	28
5.20 sort-students.h	28
5.21 include/student-app.h File Reference	28
5.21.1 Function Documentation	28
5.21.1.1 runStudentApp()	28
5.22 student-app.h	28
5.23 include/student-menu.h File Reference	29
5.23.1 Function Documentation	29
5.23.1.1 handleMenuChoice()	29
5.23.1.2 showMainMenu()	30
5.23.1.3 startMainMenu()	30
5.24 student-menu.h	30
5.25 include/student-rule-of-five-operators-test.h File Reference	30
5.25.1 Function Documentation	30
5.25.1.1 handleStudentRuleOfFiveAndOperatorsTest()	30
5.26 student-rule-of-five-operators-test.h	31
5.27 include/student.h File Reference	31
5.28 student.h	32
5.29 src/create-student.cpp File Reference	33
5.29.1 Function Documentation	33

5.29.1.1 createStudentFullyRandom()	33
5.29.1.2 createStudentManual()	34
5.29.1.3 createStudentRandomGrades()	34
5.29.1.4 createStudentsFromFile() [1/2]	34
5.29.1.5 createStudentsFromFile() [2/2]	34
5.29.1.6 randomName()	34
5.29.1.7 randomSurname()	34
5.30 src/data-sorting/sort-nuskriausti-and-protingi.cpp File Reference	34
5.30.1 Function Documentation	35
5.30.1.1 handleSortNuskriaustiAndProtingiFromFile()	35
5.30.1.2 handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > >()	35
5.30.1.3 handleSortNuskriaustiAndProtingiFromFile< std::list< Student > >()	35
5.30.1.4 handleSortNuskriaustiAndProtingiFromFile< vector< Student > >()	35
5.30.1.5 sortNuskriaustiAndProtingiFirstStrategy() [1/4]	36
5.30.1.6 sortNuskriaustiAndProtingiFirstStrategy() [2/4]	36
5.30.1.7 sortNuskriaustiAndProtingiFirstStrategy() [3/4]	36
5.30.1.8 sortNuskriaustiAndProtingiFirstStrategy() [4/4]	36
5.30.1.9 sortNuskriaustiAndProtingiSecondStrategy() [1/4]	36
5.30.1.10 sortNuskriaustiAndProtingiSecondStrategy() [2/4]	36
5.30.1.11 sortNuskriaustiAndProtingiSecondStrategy() [3/4]	36
5.30.1.12 sortNuskriaustiAndProtingiSecondStrategy() [4/4]	36
5.30.1.13 sortNuskriaustiAndProtingiThirdStrategy() [1/4]	37
5.30.1.14 sortNuskriaustiAndProtingiThirdStrategy() [2/4]	37
5.30.1.15 sortNuskriaustiAndProtingiThirdStrategy() [3/4]	37
5.30.1.16 sortNuskriaustiAndProtingiThirdStrategy() [4/4]	37
5.31 src/data-sorting/sort-students.cpp File Reference	37
5.31.1 Function Documentation	38
5.31.1.1 runSortStudentsChoicePrompt() [1/4]	38
5.31.1.2 runSortStudentsChoicePrompt() [2/4]	38
5.31.1.3 runSortStudentsChoicePrompt() [3/4]	38
5.31.1.4 runSortStudentsChoicePrompt() [4/4]	38
5.31.1.5 sortStudents() [1/4]	38
5.31.1.6 sortStudents() [2/4]	38
5.31.1.7 sortStudents() [3/4]	39
5.31.1.8 sortStudents() [4/4]	39
5.32 src/generate-student-list.cpp File Reference	39
5.32.1 Function Documentation	40
5.32.1.1 generateStudentListFile()	40
5.32.1.2 handleGenerateStudentListFile()	40
5.32.1.3 printHeader()	40
5.32.1.4 printStudentListRow()	40
5.33 src/main-menu.cpp File Reference	40

5.33.1 Function Documentation	41
5.33.1.1 handleMenuChoice()	41
5.33.1.2 handleMenuChoice< std::deque< Student > >()	41
5.33.1.3 handleMenuChoice< std::list< Student > >()	41
5.33.1.4 handleMenuChoice< vector< Student > >()	41
5.33.1.5 showMainMenu()	41
5.33.1.6 startMainMenu()	42
5.33.1.7 startMainMenu< std::deque< Student > >()	42
5.33.1.8 startMainMenu< std::list< Student > >()	42
5.33.1.9 startMainMenu< vector< Student > >()	42
5.34 src/main.cpp File Reference	42
5.34.1 Function Documentation	43
5.34.1.1 main()	43
5.35 src/output-results.cpp File Reference	43
5.35.1 Function Documentation	43
5.35.1.1 chooseOutputAndPrint() [1/4]	43
5.35.1.2 chooseOutputAndPrint() [2/4]	44
5.35.1.3 chooseOutputAndPrint() [3/4]	44
5.35.1.4 chooseOutputAndPrint() [4/4]	44
5.35.1.5 outputResults() [1/4]	44
5.35.1.6 outputResults() [2/4]	44
5.35.1.7 outputResults() [3/4]	44
5.35.1.8 outputResults() [4/4]	44
5.36 src/student.cpp File Reference	45
5.36.1 Function Documentation	45
5.36.1.1 operator<<()	45
5.36.1.2 operator>>()	45
5.37 src/test/generate-student-list-test.cpp File Reference	45
5.37.1 Function Documentation	46
5.37.1.1 handleGenerateStudentListTest()	46
5.38 src/test/output-nuskriausti-and-protingi-test.cpp File Reference	46
5.38.1 Function Documentation	47
5.38.1.1 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles()	47
5.38.1.2 outputNuskriaustiAndProtingiTestForContainer()	47
5.39 src/test/student-rule-of-five-operators-test.cpp File Reference	47
5.39.1 Function Documentation	48
5.39.1.1 handleStudentRuleOfFiveAndOperatorsTest()	48
5.39.1.2 printTestResult()	48
5.39.1.3 testCopyAssignment()	48
5.39.1.4 testCopyConstructor()	48
5.39.1.5 testDestructor()	48
5.39.1.6 testHumanIsAbstractAndNotInstantiable()	48

5.39.1.7 testInputOperatorInvalid()	48
5.39.1.8 testInputOperatorValid()	49
5.39.1.9 testMoveAssignment()	49
5.39.1.10 testMoveConstructor()	49
5.39.1.11 testOutputOperator()	49
5.40 src/utls/grading-utls.cpp File Reference	49
5.40.1 Function Documentation	50
5.40.1.1 calculateFinalGradeAverage()	50
5.40.1.2 calculateFinalGradeMedian()	50
5.40.1.3 randomGrade()	50
5.41 src/utls/input-utls.cpp File Reference	50
5.41.1 Function Documentation	51
5.41.1.1 readIntInRange()	51
5.41.1.2 readSingleIntToken()	51
5.41.1.3 readSingleStringToken()	51
5.41.1.4 readYesOrNo()	51
Index	53

Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Human	7
Student	10

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Human		7
Student		10

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

include/create-student.h	15
include/generate-student-list-test.h	17
include/generate-student-list.h	18
include/grading-utils.h	19
include/human.h	20
include/input-utils.h	21
include/output-nuskriausti-and-protingi-test.h	23
include/output-results.h	23
include/sort-nuskriausti-and-protingi.h	25
include/sort-students.h	27
include/student-app.h	28
include/student-menu.h	29
include/student-rule-of-five-operators-test.h	30
include/student.h	31
src/create-student.cpp	33
src/generate-student-list.cpp	39
src/main-menu.cpp	40
src/main.cpp	42
src/output-results.cpp	43
src/student.cpp	45
src/data-sorting/sort-nuskriausti-and-protingi.cpp	34
src/data-sorting/sort-students.cpp	37
src/test/generate-student-list-test.cpp	45
src/test/output-nuskriausti-and-protingi-test.cpp	46
src/test/student-rule-of-five-operators-test.cpp	47
src/utlis/grading-utils.cpp	49
src/utlis/input-utils.cpp	50

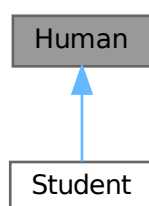
Chapter 4

Class Documentation

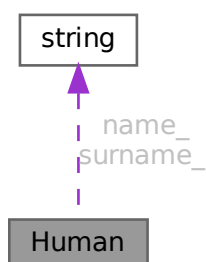
4.1 Human Class Reference

```
#include <human.h>
```

Inheritance diagram for Human:



Collaboration diagram for Human:



Public Member Functions

- [Human](#) ()=default
- [Human](#) (string [name](#), string [surname](#))
- [Human](#) (const [Human](#) &)=default
- [Human](#) ([Human](#) &&) noexcept=default
- [Human](#) & [operator=](#) (const [Human](#) &)=default
- [Human](#) & [operator=](#) ([Human](#) &&) noexcept=default
- virtual [~Human](#) ()=0
- const string & [name](#) () const
- const string & [surname](#) () const
- void [setName](#) (string [name](#))
- void [setSurname](#) (string [surname](#))

Protected Attributes

- string [name_](#)
- string [surname_](#)

4.1.1 Constructor & Destructor Documentation

4.1.1.1 [Human\(\)](#) [1/4]

```
Human::Human ( ) [default]
```

4.1.1.2 [Human\(\)](#) [2/4]

```
Human::Human (
    string name,
    string surname ) [inline]
```

4.1.1.3 [Human\(\)](#) [3/4]

```
Human::Human (
    const Human & ) [default]
```

4.1.1.4 [Human\(\)](#) [4/4]

```
Human::Human (
    Human && ) [default], [noexcept]
```

4.1.1.5 [~Human\(\)](#)

```
Human::~~Human ( ) [inline], [pure virtual], [default]
```


4.1.2 Member Function Documentation

4.1.2.1 name()

```
const string & Human::name ( ) const [inline]
```

4.1.2.2 operator=() [1/2]

```
Human & Human::operator= (
    const Human & ) [default]
```

4.1.2.3 operator=() [2/2]

```
Human & Human::operator= (
    Human && ) [default], [noexcept]
```

4.1.2.4 setName()

```
void Human::setName (
    string name ) [inline]
```

4.1.2.5 setSurname()

```
void Human::setSurname (
    string surname ) [inline]
```

4.1.2.6 surname()

```
const string & Human::surname ( ) const [inline]
```

4.1.3 Member Data Documentation

4.1.3.1 name_

```
string Human::name_ [protected]
```

4.1.3.2 surname_

```
string Human::surname_ [protected]
```

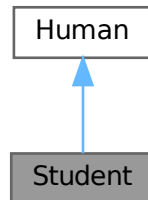
The documentation for this class was generated from the following file:

- [include/human.h](#)

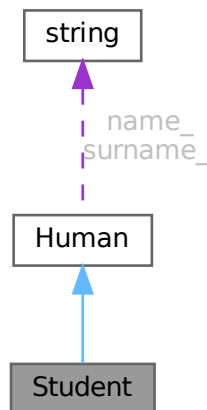
4.2 Student Class Reference

```
#include <student.h>
```

Inheritance diagram for Student:



Collaboration diagram for Student:



Public Member Functions

- [Student](#) ()
- [Student](#) (string [name](#), string [surname](#))
- [Student](#) (string [name](#), string [surname](#), vector< int > [homeworkGrades](#), int [examGrade](#))
- [Student](#) (std::istream &is)
- [Student](#) (const [Student](#) &other)
- [Student](#) ([Student](#) &&other) noexcept
- [Student](#) & [operator=](#) (const [Student](#) &other)
- [Student](#) & [operator=](#) ([Student](#) &&other) noexcept
- [~Student](#) ()

- int `examGrade` () const
- const vector< int > & `homeworkGrades` () const
- double `finalGradeAverage` () const
- double `finalGradeMedian` () const
- void `setExamGrade` (int `examGrade`)
- void `setHomeworkGrades` (vector< int > `homeworkGrades`)
- std::istream & `readStudent` (std::istream &is)
- const string & `name` () const
- const string & `surname` () const
- void `setName` (string `name`)
- void `setSurname` (string `surname`)

Public Member Functions inherited from `Human`

- `Human` ()=default
- `Human` (string `name`, string `surname`)
- `Human` (const `Human` &)=default
- `Human` (`Human` &&) noexcept=default
- `Human` & `operator=` (const `Human` &)=default
- `Human` & `operator=` (`Human` &&) noexcept=default
- virtual `~Human` ()=0
- const string & `name` () const
- const string & `surname` () const
- void `setName` (string `name`)
- void `setSurname` (string `surname`)

Friends

- std::istream & `operator>>` (std::istream &is, `Student` &student)
- std::ostream & `operator<<` (std::ostream &os, const `Student` &student)

Additional Inherited Members

Protected Attributes inherited from `Human`

- string `name_`
- string `surname_`

4.2.1 Constructor & Destructor Documentation

4.2.1.1 `Student()` [1/6]

```
Student::Student ( ) [inline]
```

4.2.1.2 `Student()` [2/6]

```
Student::Student (
    string name,
    string surname ) [inline]
```

4.2.1.3 Student() [3/6]

```
Student::Student (
    string name,
    string surname,
    vector< int > homeworkGrades,
    int examGrade ) [inline]
```

4.2.1.4 Student() [4/6]

```
Student::Student (
    std::istream & is )
```

4.2.1.5 Student() [5/6]

```
Student::Student (
    const Student & other ) [inline]
```

4.2.1.6 Student() [6/6]

```
Student::Student (
    Student && other ) [inline], [noexcept]
```

4.2.1.7 ~Student()

```
Student::~Student ( ) [inline]
```

4.2.2 Member Function Documentation

4.2.2.1 examGrade()

```
int Student::examGrade ( ) const [inline]
```

4.2.2.2 finalGradeAverage()

```
double Student::finalGradeAverage ( ) const
```

4.2.2.3 finalGradeMedian()

```
double Student::finalGradeMedian ( ) const
```

4.2.2.4 homeworkGrades()

```
const vector< int > & Student::homeworkGrades ( ) const [inline]
```

4.2.2.5 name()

```
const string & Human::name ( ) const [inline]
```

4.2.2.6 operator=() [1/2]

```
Student & Student::operator= (
    const Student & other ) [inline]
```

4.2.2.7 operator=() [2/2]

```
Student & Student::operator= (
    Student && other ) [inline], [noexcept]
```

4.2.2.8 readStudent()

```
std::istream & Student::readStudent (
    std::istream & is )
```

4.2.2.9 setExamGrade()

```
void Student::setExamGrade (
    int examGrade ) [inline]
```

4.2.2.10 setHomeworkGrades()

```
void Student::setHomeworkGrades (
    vector< int > homeworkGrades ) [inline]
```

4.2.2.11 setName()

```
void Human::setName (
    string name ) [inline]
```

4.2.2.12 setSurname()

```
void Human::setSurname (
    string surname ) [inline]
```

4.2.2.13 surname()

```
const string & Human::surname ( ) const [inline]
```

4.2.3 Friends And Related Symbol Documentation

4.2.3.1 `operator<<`

```
std::ostream & operator<< (  
    std::ostream & os,  
    const Student & student ) [friend]
```

4.2.3.2 `operator>>`

```
std::istream & operator>> (  
    std::istream & is,  
    Student & student ) [friend]
```

The documentation for this class was generated from the following files:

- [include/student.h](#)
- [src/student.cpp](#)

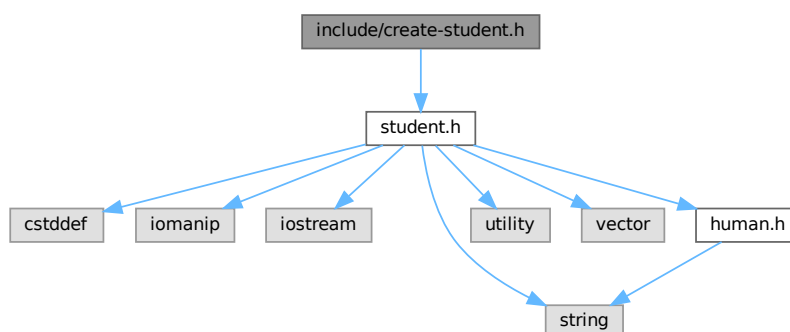
Chapter 5

File Documentation

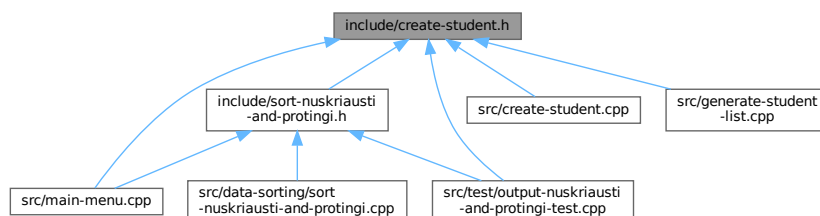
5.1 include/create-student.h File Reference

```
#include "student.h"
```

Include dependency graph for create-student.h:



This graph shows which files directly or indirectly include this file:



Functions

- string [randomName](#) ()
- string [randomSurname](#) ()
- [Student](#) [createStudentManual](#) ()
- [Student](#) [createStudentRandomGrades](#) ()
- [Student](#) [createStudentFullyRandom](#) ()
- template<typename T >
T [createStudentsFromFile](#) (const string &filename)

5.1.1 Function Documentation

5.1.1.1 [createStudentFullyRandom\(\)](#)

```
Student createStudentFullyRandom ( )
```

5.1.1.2 [createStudentManual\(\)](#)

```
Student createStudentManual ( )
```

5.1.1.3 [createStudentRandomGrades\(\)](#)

```
Student createStudentRandomGrades ( )
```

5.1.1.4 [createStudentsFromFile\(\)](#)

```
template<typename T >
T createStudentsFromFile (
    const string & filename )
```

5.1.1.5 [randomName\(\)](#)

```
string randomName ( )
```

5.1.1.6 [randomSurname\(\)](#)

```
string randomSurname ( )
```

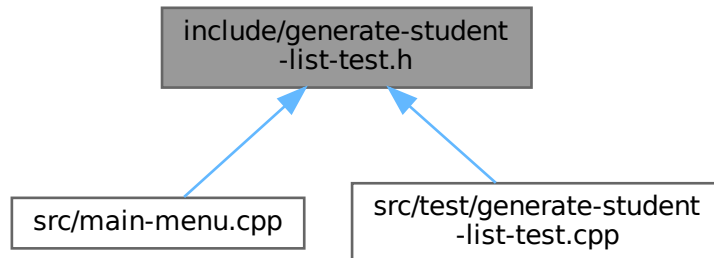
5.2 [create-student.h](#)

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 string randomName();
00006 string randomSurname();
00007
00008 Student createStudentManual();
00009 Student createStudentRandomGrades();
00010 Student createStudentFullyRandom();
00011
00012 template <typename T>
00013 T createStudentsFromFile(const string& filename);
```


5.3 include/generate-student-list-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [handleGenerateStudentListTest](#) ()

5.3.1 Function Documentation

5.3.1.1 handleGenerateStudentListTest()

```
void handleGenerateStudentListTest ( )
```

5.4 generate-student-list-test.h

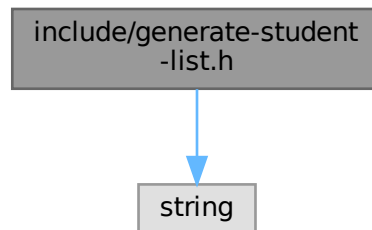
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void handleGenerateStudentListTest ();
```

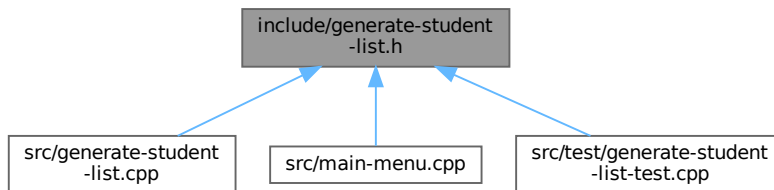
5.5 include/generate-student-list.h File Reference

```
#include <string>
```

Include dependency graph for generate-student-list.h:



This graph shows which files directly or indirectly include this file:



Functions

- void [generateStudentListFile](#) (int studentCount, const std::string &filename)
- void [handleGenerateStudentListFile](#) ()

5.5.1 Function Documentation

5.5.1.1 generateStudentListFile()

```
void generateStudentListFile (  
    int studentCount,  
    const std::string & filename )
```

5.5.1.2 handleGenerateStudentListFile()

```
void handleGenerateStudentListFile ( )
```

5.6 generate-student-list.h

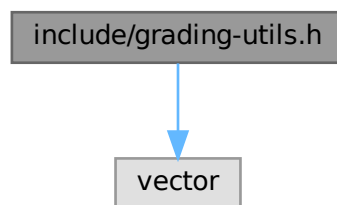
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <string>
00004
00005 void generateStudentListFile(int studentCount, const std::string& filename);
00006 void handleGenerateStudentListFile();
```

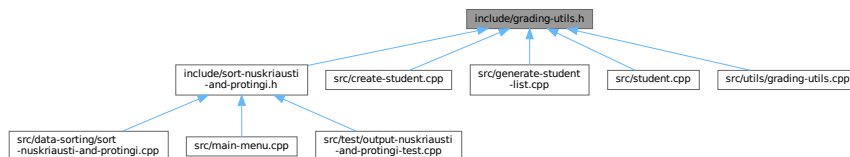
5.7 include/grading-utils.h File Reference

```
#include <vector>
```

Include dependency graph for grading-utils.h:



This graph shows which files directly or indirectly include this file:



Functions

- int [randomGrade](#) (int min=1, int max=10)
- double [calculateFinalGradeAverage](#) (const vector< int > &homeworkGrades, int examGrade)
- double [calculateFinalGradeMedian](#) (const vector< int > &homeworkGrades, int examGrade)

5.7.1 Function Documentation

5.7.1.1 calculateFinalGradeAverage()

```
double calculateFinalGradeAverage (
    const vector< int > & homeworkGrades,
    int examGrade )
```

5.7.1.2 calculateFinalGradeMedian()

```
double calculateFinalGradeMedian (
    const vector< int > & homeworkGrades,
    int examGrade )
```

5.7.1.3 randomGrade()

```
int randomGrade (
    int min = 1,
    int max = 10 )
```

5.8 grading-utils.h

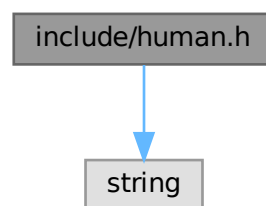
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <vector>
00004
00005 using std::vector;
00006
00007 int randomGrade(int min = 1, int max = 10);
00008 double calculateFinalGradeAverage(const vector<int>& homeworkGrades, int examGrade);
00009 double calculateFinalGradeMedian(const vector<int>& homeworkGrades, int examGrade);
```

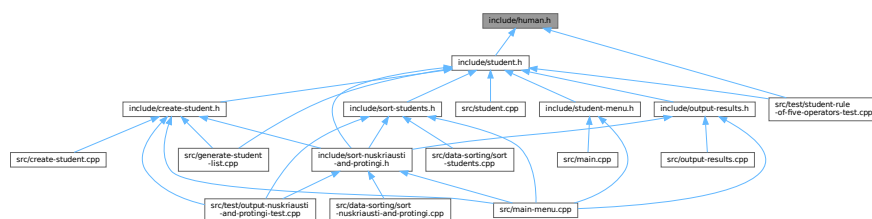
5.9 include/human.h File Reference

```
#include <string>
```

Include dependency graph for human.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Human](#)

5.10 human.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include <string>
00004
00005 using std::string;
00006
00007 class Human {
00008 protected:
00009     string name_;
00010     string surname_;
00011
00012 public:
00013     Human() = default;
00014     Human(string name, string surname) : name_(name), surname_(surname) {}
00015     Human(const Human&) = default;
00016     Human(Human&&) noexcept = default;
00017     Human& operator=(const Human&) = default;
00018     Human& operator=(Human&&) noexcept = default;
00019     virtual ~Human() = 0;
00020
00021     inline const string& name() const { return name_; }
00022     inline const string& surname() const { return surname_; }
00023
00024     inline void setName(string name) { name_ = name; }
00025     inline void setSurname(string surname) { surname_ = surname; }
00026 };
00027
00028 inline Human::~Human() = default;

```

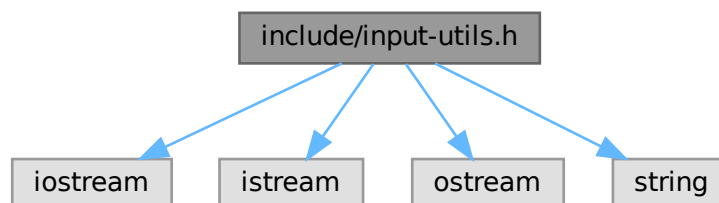
5.11 include/input-utils.h File Reference

```

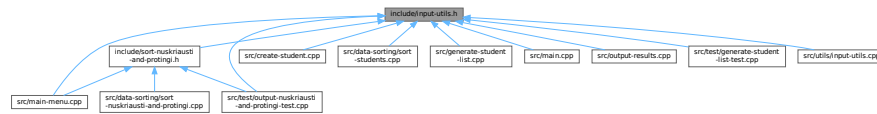
#include <iostream>
#include <istream>
#include <ostream>
#include <string>

```

Include dependency graph for input-utils.h:



This graph shows which files directly or indirectly include this file:



Functions

- int [readIntInRange](#) (const string &prompt, int min, int max)
- string [readSingleStringToken](#) (const string &prompt)
- char [readYesOrNo](#) (const string &prompt)
- int [readSingleIntToken](#) (const string &prompt)

5.11.1 Function Documentation

5.11.1.1 readIntInRange()

```
int readIntInRange (
    const string & prompt,
    int min,
    int max )
```

5.11.1.2 readSingleIntToken()

```
int readSingleIntToken (
    const string & prompt )
```

5.11.1.3 readSingleStringToken()

```
string readSingleStringToken (
    const string & prompt )
```

5.11.1.4 readYesOrNo()

```
char readYesOrNo (
    const string & prompt )
```

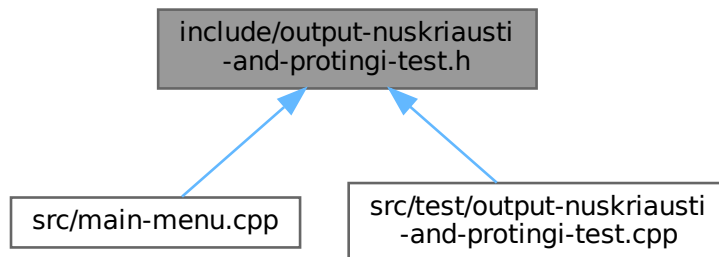
5.12 input-utils.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <iostream>
00004 #include <istream>
00005 #include <ostream>
00006 #include <string>
00007
00008 using std::istream;
00009 using std::ostream;
00010 using std::string;
00011
00012 int readIntInRange(const string& prompt, int min, int max);
00013 string readSingleStringToken(const string& prompt);
00014 char readYesOrNo(const string& prompt);
00015 int readSingleIntToken(const string& prompt);
```

5.13 include/output-nuskriausti-and-protingi-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles\(\)](#)

5.13.1 Function Documentation

5.13.1.1 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles()

```
void outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles ( )
```

5.14 output-nuskriausti-and-protingi-test.h

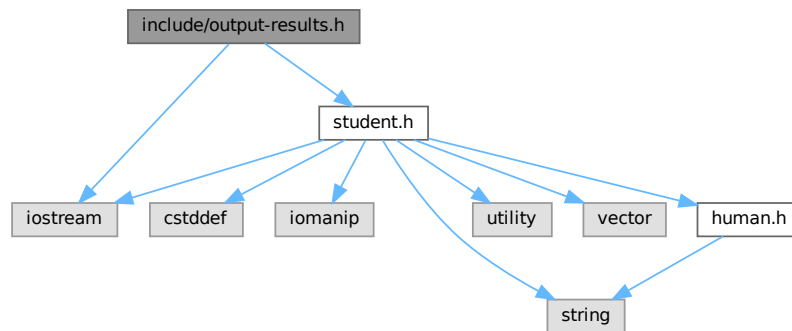
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles();
```

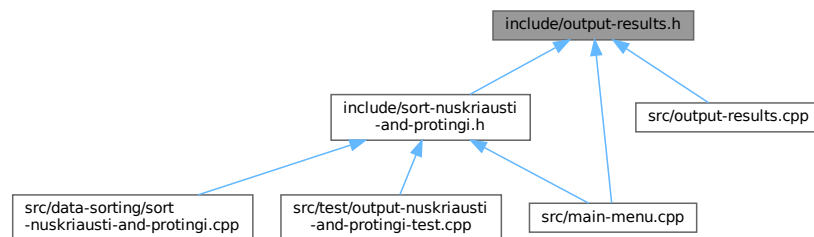
5.15 include/output-results.h File Reference

```
#include "student.h"
#include <iostream>
```

Include dependency graph for output-results.h:



This graph shows which files directly or indirectly include this file:



Functions

- `template<typename T >`
void `outputResults` (const T &students, std::ostream &out)
- `template<typename T >`
void `chooseOutputAndPrint` (T &students)

5.15.1 Function Documentation

5.15.1.1 chooseOutputAndPrint()

```

template<typename T >
void chooseOutputAndPrint (
    T & students )

```

5.15.1.2 outputResults()

```

template<typename T >
void outputResults (
    const T & students,
    std::ostream & out )

```


5.16 output-results.h

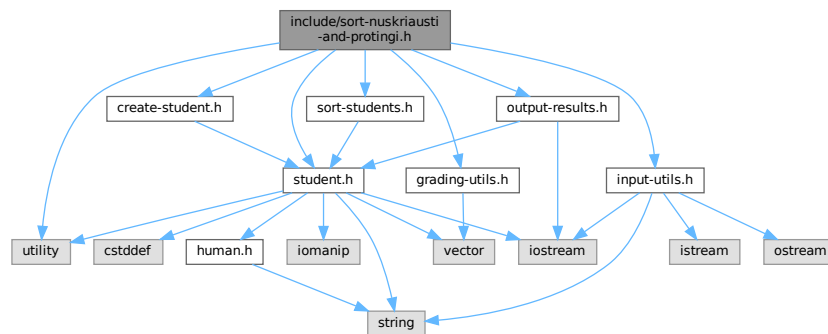
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 #include <iostream>
00006
00007 template <typename T>
00008 void outputResults(const T& students, std::ostream& out);
00009
00010 template <typename T>
00011 void chooseOutputAndPrint(T& students);
```

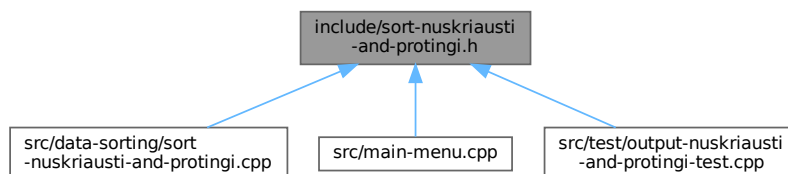
5.17 include/sort-nuskriausti-and-protingi.h File Reference

```
#include "create-student.h"
#include "input-utils.h"
#include "output-results.h"
#include "sort-students.h"
#include "student.h"
#include "grading-utils.h"
#include <utility>
```

Include dependency graph for sort-nuskriausti-and-protingi.h:



This graph shows which files directly or indirectly include this file:



Functions

- `template<typename T >`
`void handleSortNuskriaustiAndProtingiFromFile ( )`
- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (const T &students)`
- `template<typename T >`
`T sortNuskriaustiAndProtingiSecondStrategy (T &students)`
- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (T &students)`

5.17.1 Function Documentation

5.17.1.1 [handleSortNuskriaustiAndProtingiFromFile\(\)](#)

```
template<typename T >
void handleSortNuskriaustiAndProtingiFromFile ( )
```

5.17.1.2 [sortNuskriaustiAndProtingiFirstStrategy\(\)](#)

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (
    const T & students )
```

5.17.1.3 [sortNuskriaustiAndProtingiSecondStrategy\(\)](#)

```
template<typename T >
T sortNuskriaustiAndProtingiSecondStrategy (
    T & students )
```

5.17.1.4 [sortNuskriaustiAndProtingiThirdStrategy\(\)](#)

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (
    T & students )
```

5.18 [sort-nuskriausti-and-protingi.h](#)

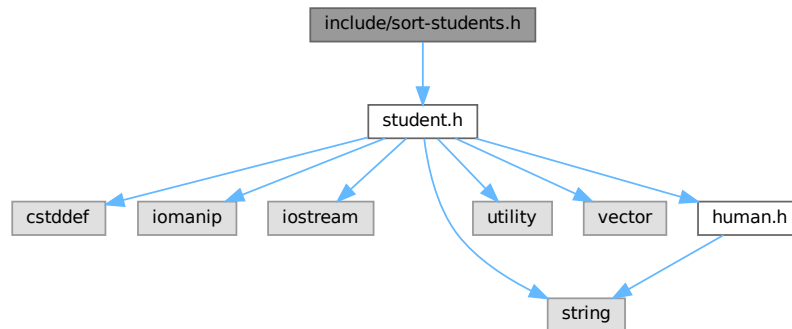
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "create-student.h"
00004 #include "input-utils.h"
00005 #include "output-results.h"
00006 #include "sort-students.h"
00007 #include "student.h"
00008 #include "grading-utils.h"
00009
00010 #include <utility>
00011
00012 template <typename T>
00013 void handleSortNuskriaustiAndProtingiFromFile();
00014
00015 template <typename T>
00016 std::pair<T, T> sortNuskriaustiAndProtingiFirstStrategy(const T& students);
00017
00018 template <typename T>
00019 T sortNuskriaustiAndProtingiSecondStrategy(T& students);
00020
00021 template <typename T>
00022 std::pair<T, T> sortNuskriaustiAndProtingiThirdStrategy(T& students);
```

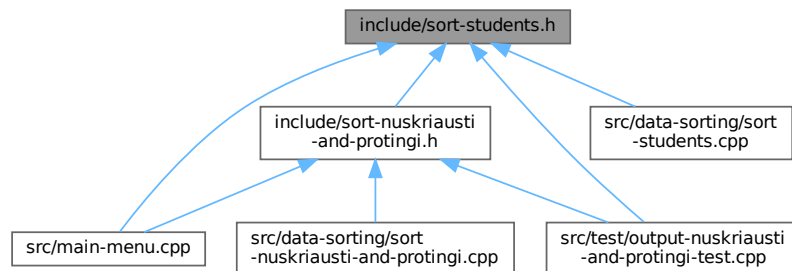
5.19 include/sort-students.h File Reference

```
#include "student.h"
```

Include dependency graph for sort-students.h:



This graph shows which files directly or indirectly include this file:



Functions

- `template<typename T >`
void `runSortStudentsChoicePrompt` (T &students)
- `template<typename T >`
void `sortStudents` (T &students, int sortChoice, bool ascending)

5.19.1 Function Documentation

5.19.1.1 runSortStudentsChoicePrompt()

```
template<typename T >
void runSortStudentsChoicePrompt (
    T & students )
```

5.19.1.2 sortStudents()

```
template<typename T >
void sortStudents (
    T & students,
    int sortChoice,
    bool ascending )
```

5.20 sort-students.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 template <typename T>
00006 void runSortStudentsChoicePrompt (T& students);
00007
00008 template <typename T>
00009 void sortStudents (T& students, int sortChoice, bool ascending);
```

5.21 include/student-app.h File Reference

Functions

- void [runStudentApp](#) ()

5.21.1 Function Documentation

5.21.1.1 runStudentApp()

```
void runStudentApp ( )
```

5.22 student-app.h

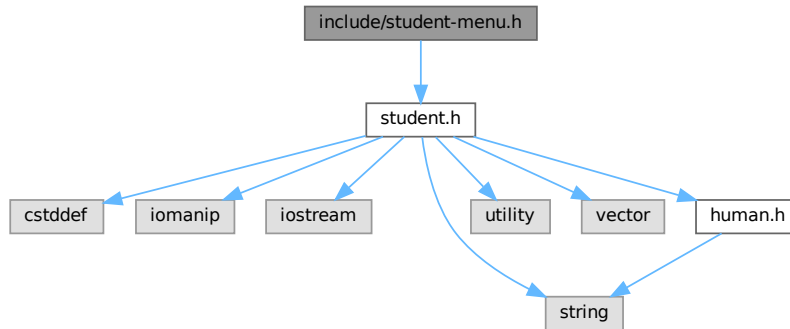
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void runStudentApp ();
```

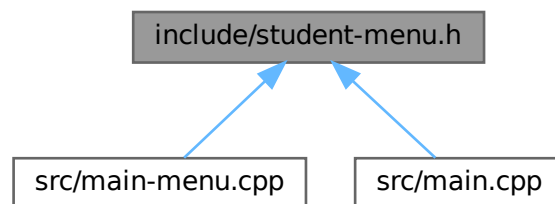
5.23 include/student-menu.h File Reference

```
#include "student.h"
```

Include dependency graph for student-menu.h:



This graph shows which files directly or indirectly include this file:



Functions

- void [showMainMenu](#) ()
- template<typename T >
void [handleMenuChoice](#) (int choice, T &students)
- template<typename T >
void [startMainMenu](#) ()

5.23.1 Function Documentation

5.23.1.1 handleMenuChoice()

```
template<typename T >
void handleMenuChoice (
    int choice,
    T & students )
```

5.23.1.2 showMainMenu()

```
void showMainMenu ( )
```

5.23.1.3 startMainMenu()

```
template<typename T >
void startMainMenu ( )
```

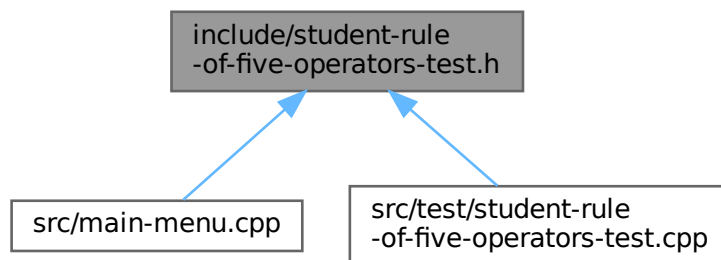
5.24 student-menu.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 void showMainMenu();
00006
00007 template <typename T>
00008 void handleMenuChoice(int choice, T& students);
00009
00010 template <typename T>
00011 void startMainMenu();
```

5.25 include/student-rule-of-five-operators-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [handleStudentRuleOfFiveAndOperatorsTest](#) ()

5.25.1 Function Documentation

5.25.1.1 handleStudentRuleOfFiveAndOperatorsTest()

```
void handleStudentRuleOfFiveAndOperatorsTest ( )
```

5.26 student-rule-of-five-operators-test.h

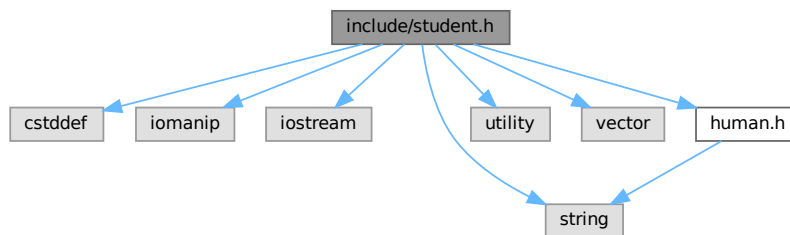
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void handleStudentRuleOfFiveAndOperatorsTest ();
```

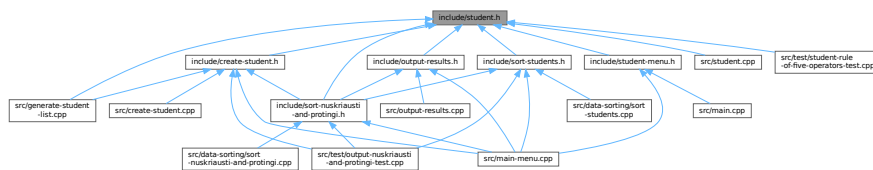
5.27 include/student.h File Reference

```
#include <cstdint>
#include <iomanip>
#include <iostream>
#include <string>
#include <utility>
#include <vector>
#include "human.h"
```

Include dependency graph for student.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Student](#)

5.28 student.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include <cstdint>
00004 #include <iomanip>
00005 #include <iostream>
00006 #include <string>
00007 #include <utility>
00008 #include <vector>
00009
00010 #include "human.h"
00011
00012 using std::string;
00013 using std::vector;
00014 using std::cout;
00015 using std::cin;
00016 using std::endl;
00017 using std::setw;
00018 using std::left;
00019 using std::right;
00020
00021 class Student : public Human {
00022 private:
00023     int examGrade_;
00024     vector<int> homeworkGrades_;
00025
00026 public:
00027     Student() : Human(), examGrade_(0) {}
00028
00029     Student(string name, string surname) : Human(name, surname), examGrade_(0) {}
00030     Student(string name, string surname, vector<int> homeworkGrades, int examGrade) : Human(name,
00031 surname), examGrade_(examGrade), homeworkGrades_(homeworkGrades) {}
00032     Student(std::istream& is);
00033     Student(const Student& other)
00034         : Human(other),
00035           examGrade_(other.examGrade_),
00036           homeworkGrades_(other.homeworkGrades_) {}
00037
00038     Student(Student&& other) noexcept
00039         : Human(std::move(other)),
00040           examGrade_(other.examGrade_),
00041           homeworkGrades_(std::move(other.homeworkGrades_)) {
00042         other.examGrade_ = 0;
00043     }
00044     Student& operator=(const Student& other) {
00045         if (this == &other) {
00046             return *this;
00047         }
00048         Human::operator=(other);
00049         examGrade_ = other.examGrade_;
00050         homeworkGrades_ = other.homeworkGrades_;
00051
00052         return *this;
00053     }
00054     Student& operator=(Student&& other) noexcept {
00055         if (this == &other) {
00056             return *this;
00057         }
00058         Human::operator=(std::move(other));
00059         examGrade_ = other.examGrade_;
00060         homeworkGrades_ = std::move(other.homeworkGrades_);
00061         other.examGrade_ = 0;
00062
00063         return *this;
00064     }
00065     ~Student() {
00066         name_.clear();
00067         surname_.clear();
00068         homeworkGrades_.clear();
00069         examGrade_ = 0;
00070     }
00071
00072
00073     using Human::name;
00074     using Human::surname;
00075     inline int examGrade() const { return examGrade_; }
00076     inline const vector<int>& homeworkGrades() const { return homeworkGrades_; }
00077     double finalGradeAverage() const;
00078     double finalGradeMedian() const;
00079
00080     using Human::setName;
00081     using Human::setSurname;

```



```

00082     inline void setExamGrade(int examGrade) { examGrade_ = examGrade; }
00083     inline void setHomeworkGrades(vector<int> homeworkGrades) { homeworkGrades_ = homeworkGrades; }
00084
00085     std::istream& readStudent(std::istream& is);
00086     friend std::istream& operator>>(std::istream& is, Student& student);
00087     friend std::ostream& operator<<(std::ostream& os, const Student& student);
00088 };

```

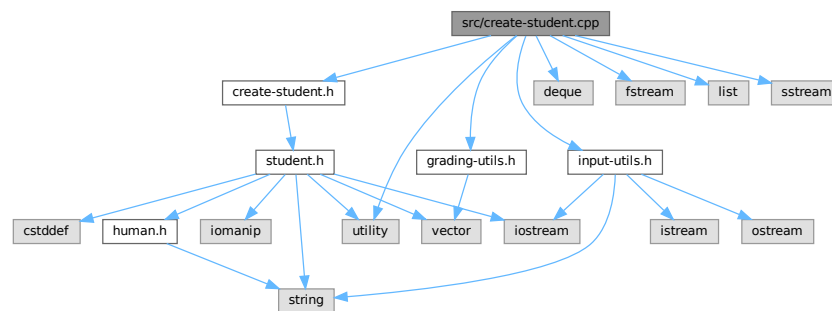
5.29 src/create-student.cpp File Reference

```

#include "create-student.h"
#include "input-utils.h"
#include "grading-utils.h"
#include <deque>
#include <fstream>
#include <list>
#include <sstream>
#include <utility>

```

Include dependency graph for create-student.cpp:



Functions

- [Student createStudentManual](#) ()
- [Student createStudentRandomGrades](#) ()
- string [randomName](#) ()
- string [randomSurname](#) ()
- [Student createStudentFullyRandom](#) ()
- template<typename T>
T [createStudentsFromFile](#) (const string &filename)
- template vector< [Student](#) > [createStudentsFromFile](#) (const string &filename)

5.29.1 Function Documentation

5.29.1.1 createStudentFullyRandom()

```
Student createStudentFullyRandom ( )
```

5.29.1.2 createStudentManual()

```
Student createStudentManual ( )
```

5.29.1.3 createStudentRandomGrades()

```
Student createStudentRandomGrades ( )
```

5.29.1.4 createStudentsFromFile() [1/2]

```
template<typename T >
template std::deque< Student > createStudentsFromFile (
    const string & filename )
```

5.29.1.5 createStudentsFromFile() [2/2]

```
template vector< Student > createStudentsFromFile (
    const string & filename )
```

5.29.1.6 randomName()

```
string randomName ( )
```

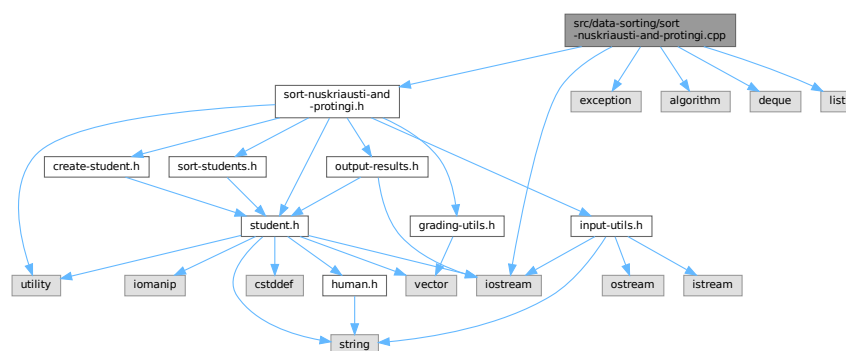
5.29.1.7 randomSurname()

```
string randomSurname ( )
```

5.30 src/data-sorting/sort-nuskriausti-and-protingi.cpp File Reference

```
#include "sort-nuskriausti-and-protingi.h"
#include <exception>
#include <iostream>
#include <algorithm>
#include <deque>
#include <list>
```

Include dependency graph for sort-nuskriausti-and-protingi.cpp:



Functions

- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (const T &students)`
- `template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiFirstStrategy (const vector< Student > &)`
- `template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiFirstStrategy (const std::list< Student > &)`
- `template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiFirstStrategy (const std::deque< Student > &)`
- `template<typename T >`
`T sortNuskriaustiAndProtingiSecondStrategy (T &students)`
- `template vector< Student > sortNuskriaustiAndProtingiSecondStrategy (vector< Student > &)`
- `template std::list< Student > sortNuskriaustiAndProtingiSecondStrategy (std::list< Student > &)`
- `template std::deque< Student > sortNuskriaustiAndProtingiSecondStrategy (std::deque< Student > &)`
- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (T &students)`
- `template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiThirdStrategy (vector< Student > &)`
- `template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiThirdStrategy (std::list< Student > &)`
- `template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiThirdStrategy (std::deque< Student > &)`
- `template<typename T >`
`void handleSortNuskriaustiAndProtingiFromFile ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< vector< Student > > ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< std::list< Student > > ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > > ()`

5.30.1 Function Documentation

5.30.1.1 handleSortNuskriaustiAndProtingiFromFile()

```
template<typename T >
void handleSortNuskriaustiAndProtingiFromFile ( )
```

5.30.1.2 handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > > ( )
```

5.30.1.3 handleSortNuskriaustiAndProtingiFromFile< std::list< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< std::list< Student > > ( )
```

5.30.1.4 handleSortNuskriaustiAndProtingiFromFile< vector< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< vector< Student > > ( )
```

5.30.1.5 sortNuskriaustiAndProtingiFirstStrategy() [1/4]

```
template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const std::deque< Student > & )
```

5.30.1.6 sortNuskriaustiAndProtingiFirstStrategy() [2/4]

```
template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const std::list< Student > & )
```

5.30.1.7 sortNuskriaustiAndProtingiFirstStrategy() [3/4]

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (
    const T & students )
```

5.30.1.8 sortNuskriaustiAndProtingiFirstStrategy() [4/4]

```
template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const vector< Student > & )
```

5.30.1.9 sortNuskriaustiAndProtingiSecondStrategy() [1/4]

```
template std::deque< Student > sortNuskriaustiAndProtingiSecondStrategy (
    std::deque< Student > & )
```

5.30.1.10 sortNuskriaustiAndProtingiSecondStrategy() [2/4]

```
template std::list< Student > sortNuskriaustiAndProtingiSecondStrategy (
    std::list< Student > & )
```

5.30.1.11 sortNuskriaustiAndProtingiSecondStrategy() [3/4]

```
template<typename T >
T sortNuskriaustiAndProtingiSecondStrategy (
    T & students )
```

5.30.1.12 sortNuskriaustiAndProtingiSecondStrategy() [4/4]

```
template vector< Student > sortNuskriaustiAndProtingiSecondStrategy (
    vector< Student > & )
```

5.30.1.13 sortNuskriaustiAndProtingiThirdStrategy() [1/4]

```
template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    std::deque< Student > & )
```

5.30.1.14 sortNuskriaustiAndProtingiThirdStrategy() [2/4]

```
template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    std::list< Student > & )
```

5.30.1.15 sortNuskriaustiAndProtingiThirdStrategy() [3/4]

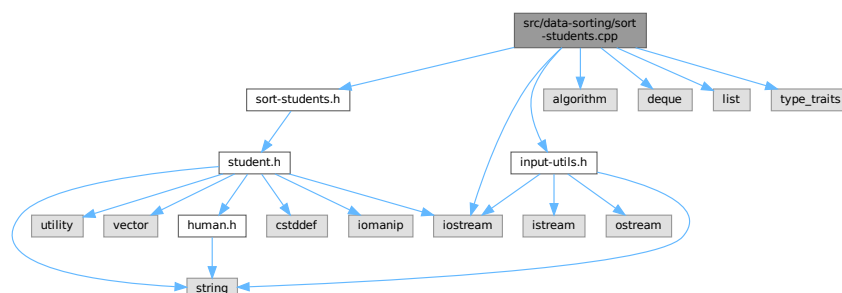
```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (
    T & students )
```

5.30.1.16 sortNuskriaustiAndProtingiThirdStrategy() [4/4]

```
template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    vector< Student > & )
```

5.31 src/data-sorting/sort-students.cpp File Reference

```
#include "sort-students.h"
#include "input-utils.h"
#include <algorithm>
#include <iostream>
#include <deque>
#include <list>
#include <type_traits>
Include dependency graph for sort-students.cpp:
```



Functions

- `template<typename T >`
void `sortStudents` (T &students, int sortChoice, bool ascending)
- `template void sortStudents` (vector< `Student` > &, int, bool)
- `template void sortStudents` (std::list< `Student` > &, int, bool)
- `template void sortStudents` (std::deque< `Student` > &, int, bool)
- `template<typename T >`
void `runSortStudentsChoicePrompt` (T &students)
- `template void runSortStudentsChoicePrompt` (vector< `Student` > &)
- `template void runSortStudentsChoicePrompt` (std::list< `Student` > &)
- `template void runSortStudentsChoicePrompt` (std::deque< `Student` > &)

5.31.1 Function Documentation

5.31.1.1 `runSortStudentsChoicePrompt()` [1/4]

```
template void runSortStudentsChoicePrompt (
    std::deque< Student > & )
```

5.31.1.2 `runSortStudentsChoicePrompt()` [2/4]

```
template void runSortStudentsChoicePrompt (
    std::list< Student > & )
```

5.31.1.3 `runSortStudentsChoicePrompt()` [3/4]

```
template<typename T >
void runSortStudentsChoicePrompt (
    T & students )
```

5.31.1.4 `runSortStudentsChoicePrompt()` [4/4]

```
template void runSortStudentsChoicePrompt (
    vector< Student > & )
```

5.31.1.5 `sortStudents()` [1/4]

```
template void sortStudents (
    std::deque< Student > & ,
    int ,
    bool )
```

5.31.1.6 `sortStudents()` [2/4]

```
template void sortStudents (
    std::list< Student > & ,
    int ,
    bool )
```

5.31.1.7 sortStudents() [3/4]

```
template<typename T >
void sortStudents (
    T & students,
    int sortChoice,
    bool ascending )
```

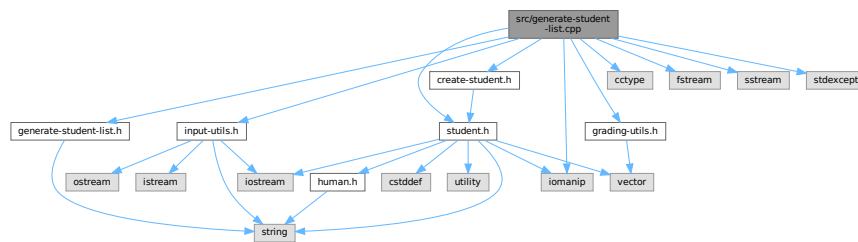
5.31.1.8 sortStudents() [4/4]

```
template void sortStudents (
    vector< Student > & ,
    int ,
    bool )
```

5.32 src/generate-student-list.cpp File Reference

```
#include "generate-student-list.h"
#include "input-utils.h"
#include "student.h"
#include "create-student.h"
#include "grading-utils.h"
#include <cctype>
#include <fstream>
#include <iomanip>
#include <sstream>
#include <stdexcept>
```

Include dependency graph for generate-student-list.cpp:



Functions

- void [printHeader](#) (std::ostream &output)
- void [printStudentListRow](#) (std::ostream &output, const [Student](#) &s)
- void [generateStudentListFile](#) (int studentCount, const std::string &filename)
- void [handleGenerateStudentListFile](#) ()

5.32.1 Function Documentation

5.32.1.1 generateStudentListFile()

```
void generateStudentListFile (
    int studentCount,
    const std::string & filename )
```

5.32.1.2 handleGenerateStudentListFile()

```
void handleGenerateStudentListFile ( )
```

5.32.1.3 printHeader()

```
void printHeader (
    std::ostream & output )
```

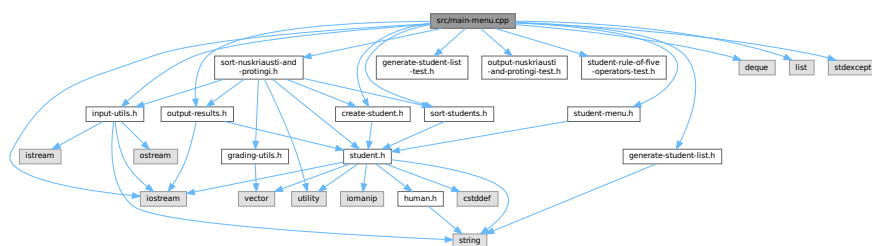
5.32.1.4 printStudentListRow()

```
void printStudentListRow (
    std::ostream & output,
    const Student & s )
```

5.33 src/main-menu.cpp File Reference

```
#include "student-menu.h"
#include "create-student.h"
#include "input-utils.h"
#include "output-results.h"
#include "generate-student-list.h"
#include "generate-student-list-test.h"
#include "output-nuskriausti-and-protingi-test.h"
#include "student-rule-of-five-operators-test.h"
#include "sort-nuskriausti-and-protingi.h"
#include "sort-students.h"
#include <iostream>
#include <deque>
#include <list>
#include <stdexcept>
```

Include dependency graph for main-menu.cpp:



Functions

- void `showMainMenu ()`
- template<typename T >
void `handleMenuChoice` (int choice, T &students)
- template void `handleMenuChoice`< vector< Student > > (int, vector< Student > &)
- template void `handleMenuChoice`< std::list< Student > > (int, std::list< Student > &)
- template void `handleMenuChoice`< std::deque< Student > > (int, std::deque< Student > &)
- template<typename T >
void `startMainMenu ()`
- template void `startMainMenu`< vector< Student > > ()
- template void `startMainMenu`< std::list< Student > > ()
- template void `startMainMenu`< std::deque< Student > > ()

5.33.1 Function Documentation

5.33.1.1 `handleMenuChoice()`

```
template<typename T >
void handleMenuChoice (
    int choice,
    T & students )
```

5.33.1.2 `handleMenuChoice< std::deque< Student > >()`

```
template void handleMenuChoice< std::deque< Student > > (
    int ,
    std::deque< Student > & )
```

5.33.1.3 `handleMenuChoice< std::list< Student > >()`

```
template void handleMenuChoice< std::list< Student > > (
    int ,
    std::list< Student > & )
```

5.33.1.4 `handleMenuChoice< vector< Student > >()`

```
template void handleMenuChoice< vector< Student > > (
    int ,
    vector< Student > & )
```

5.33.1.5 `showMainMenu()`

```
void showMainMenu ( )
```

5.33.1.6 startMainMenu()

```
template<typename T >
void startMainMenu ( )
```

5.33.1.7 startMainMenu< std::deque< Student > >()

```
template void startMainMenu< std::deque< Student > > ( )
```

5.33.1.8 startMainMenu< std::list< Student > >()

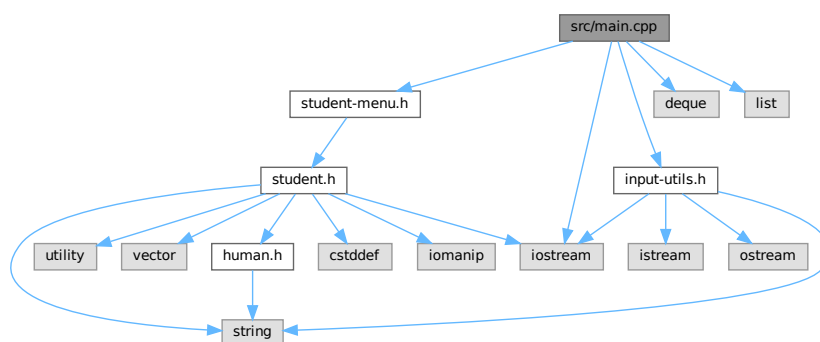
```
template void startMainMenu< std::list< Student > > ( )
```

5.33.1.9 startMainMenu< vector< Student > >()

```
template void startMainMenu< vector< Student > > ( )
```

5.34 src/main.cpp File Reference

```
#include "student-menu.h"
#include "input-utils.h"
#include <deque>
#include <iostream>
#include <list>
Include dependency graph for main.cpp:
```



Functions

- int `main` ()

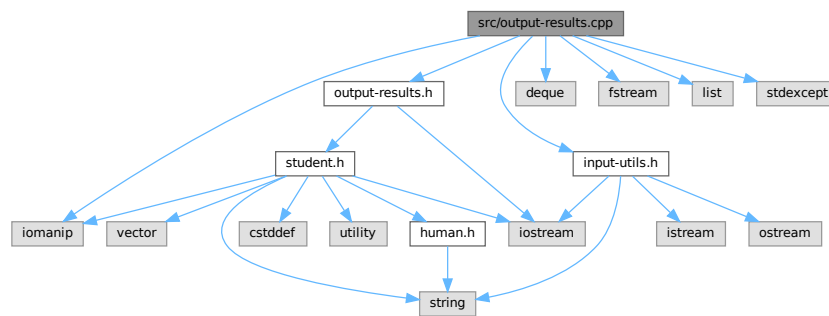
5.34.1 Function Documentation

5.34.1.1 main()

```
int main ( )
```

5.35 src/output-results.cpp File Reference

```
#include "output-results.h"
#include "input-utils.h"
#include <deque>
#include <fstream>
#include <iomanip>
#include <list>
#include <stdexcept>
Include dependency graph for output-results.cpp:
```



Functions

- `template<typename T >`
void `outputResults` (const T &students, std::ostream &out)
- `template void outputResults` (const vector< `Student` > &students, std::ostream &out)
- `template void outputResults` (const std::list< `Student` > &students, std::ostream &out)
- `template void outputResults` (const std::deque< `Student` > &students, std::ostream &out)
- `template<typename T >`
void `chooseOutputAndPrint` (T &students)
- `template void chooseOutputAndPrint` (vector< `Student` > &students)
- `template void chooseOutputAndPrint` (std::list< `Student` > &students)
- `template void chooseOutputAndPrint` (std::deque< `Student` > &students)

5.35.1 Function Documentation

5.35.1.1 chooseOutputAndPrint() [1/4]

```
template void chooseOutputAndPrint (
    std::deque< Student > & students )
```

5.35.1.2 chooseOutputAndPrint() [2/4]

```
template void chooseOutputAndPrint (
    std::list< Student > & students )
```

5.35.1.3 chooseOutputAndPrint() [3/4]

```
template<typename T >
void chooseOutputAndPrint (
    T & students )
```

5.35.1.4 chooseOutputAndPrint() [4/4]

```
template void chooseOutputAndPrint (
    vector< Student > & students )
```

5.35.1.5 outputResults() [1/4]

```
template void outputResults (
    const std::deque< Student > & students,
    std::ostream & out )
```

5.35.1.6 outputResults() [2/4]

```
template void outputResults (
    const std::list< Student > & students,
    std::ostream & out )
```

5.35.1.7 outputResults() [3/4]

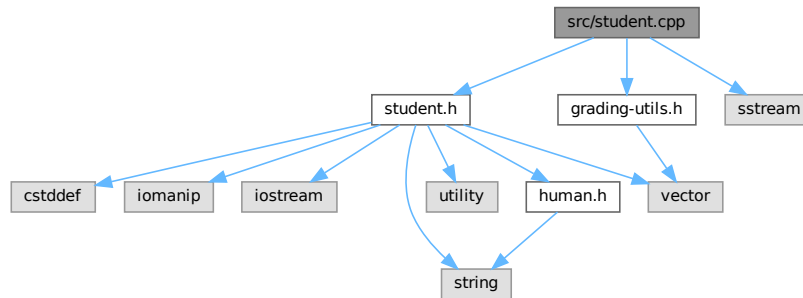
```
template<typename T >
void outputResults (
    const T & students,
    std::ostream & out )
```

5.35.1.8 outputResults() [4/4]

```
template void outputResults (
    const vector< Student > & students,
    std::ostream & out )
```

5.36 src/student.cpp File Reference

```
#include "student.h"
#include "grading-utils.h"
#include <sstream>
Include dependency graph for student.cpp:
```



Functions

- `std::istream & operator>> (std::istream &is, Student &student)`
- `std::ostream & operator<< (std::ostream &os, const Student &student)`

5.36.1 Function Documentation

5.36.1.1 operator<<()

```
std::ostream & operator<< (
    std::ostream & os,
    const Student & student )
```

5.36.1.2 operator>>()

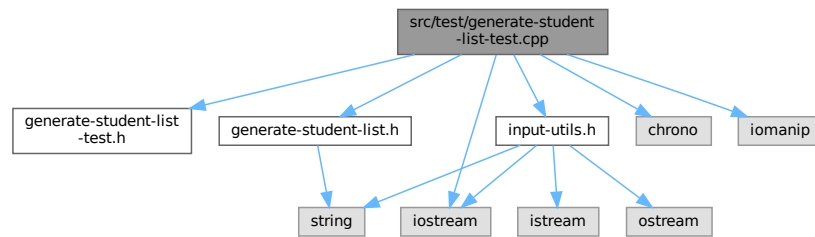
```
std::istream & operator>> (
    std::istream & is,
    Student & student )
```

5.37 src/test/generate-student-list-test.cpp File Reference

```
#include "generate-student-list-test.h"
#include "generate-student-list.h"
#include "input-utils.h"
#include <chrono>
#include <iomanip>
```

```
#include <iostream>
```

Include dependency graph for generate-student-list-test.cpp:



Functions

- void [handleGenerateStudentListTest](#) ()

5.37.1 Function Documentation

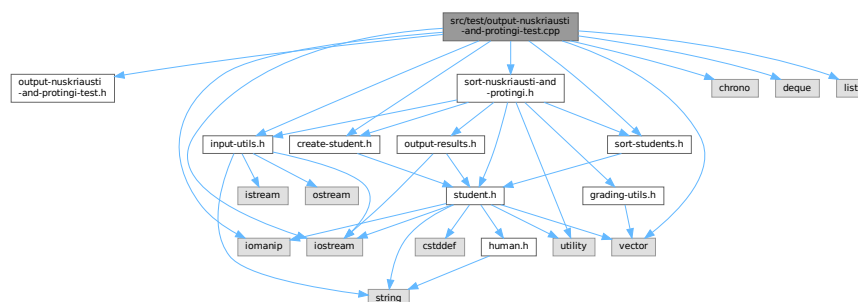
5.37.1.1 handleGenerateStudentListTest()

```
void handleGenerateStudentListTest ( )
```

5.38 src/test/output-nuskriausti-and-protingi-test.cpp File Reference

```
#include "output-nuskriausti-and-protingi-test.h"
#include "input-utils.h"
#include "create-student.h"
#include "sort-students.h"
#include "sort-nuskriausti-and-protingi.h"
#include <chrono>
#include <deque>
#include <iomanip>
#include <iostream>
#include <list>
#include <vector>
```

Include dependency graph for output-nuskriausti-and-protingi-test.cpp:



Functions

- template<typename Container >
void [outputNuskriaustiAndProtingiTestForContainer](#) (const string &filename, int strategyChoice)
- void [outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles](#) ()

5.38.1 Function Documentation

5.38.1.1 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles()

```
void outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles ( )
```

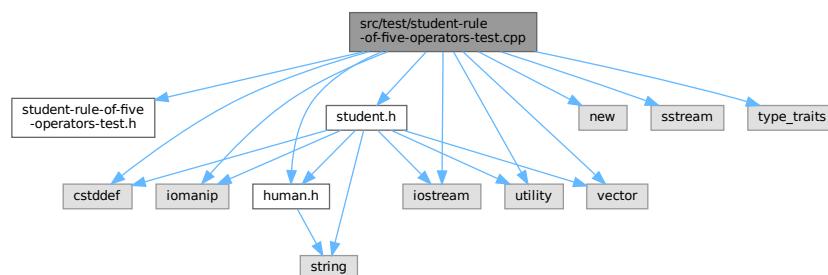
5.38.1.2 outputNuskriaustiAndProtingiTestForContainer()

```
template<typename Container >  
void outputNuskriaustiAndProtingiTestForContainer (  
    const string & filename,  
    int strategyChoice )
```

5.39 src/test/student-rule-of-five-operators-test.cpp File Reference

```
#include "student-rule-of-five-operators-test.h"  
#include "human.h"  
#include "student.h"  
#include <cstddef>  
#include <iomanip>  
#include <iostream>  
#include <new>  
#include <sstream>  
#include <type_traits>  
#include <utility>  
#include <vector>
```

Include dependency graph for student-rule-of-five-operators-test.cpp:



Functions

- void [printTestResult](#) (const char *testName, bool passed)
- bool [testCopyConstructor](#) ()
- bool [testMoveConstructor](#) ()
- bool [testCopyAssignment](#) ()
- bool [testMoveAssignment](#) ()
- bool [testDestructor](#) ()
- bool [testInputOperatorValid](#) ()
- bool [testInputOperatorInvalid](#) ()
- bool [testOutputOperator](#) ()
- bool [testHumanIsAbstractAndNotInstantiable](#) ()
- void [handleStudentRuleOfFiveAndOperatorsTest](#) ()

5.39.1 Function Documentation

5.39.1.1 [handleStudentRuleOfFiveAndOperatorsTest\(\)](#)

```
void handleStudentRuleOfFiveAndOperatorsTest ( )
```

5.39.1.2 [printTestResult\(\)](#)

```
void printTestResult (
    const char * testName,
    bool passed )
```

5.39.1.3 [testCopyAssignment\(\)](#)

```
bool testCopyAssignment ( )
```

5.39.1.4 [testCopyConstructor\(\)](#)

```
bool testCopyConstructor ( )
```

5.39.1.5 [testDestructor\(\)](#)

```
bool testDestructor ( )
```

5.39.1.6 [testHumanIsAbstractAndNotInstantiable\(\)](#)

```
bool testHumanIsAbstractAndNotInstantiable ( )
```

5.39.1.7 [testInputOperatorInvalid\(\)](#)

```
bool testInputOperatorInvalid ( )
```


5.39.1.8 testInputOperatorValid()

```
bool testInputOperatorValid ( )
```

5.39.1.9 testMoveAssignment()

```
bool testMoveAssignment ( )
```

5.39.1.10 testMoveConstructor()

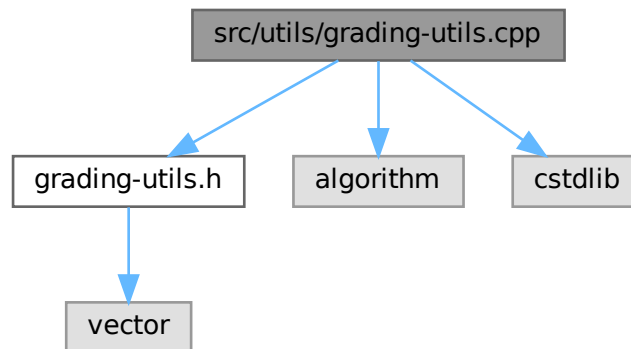
```
bool testMoveConstructor ( )
```

5.39.1.11 testOutputOperator()

```
bool testOutputOperator ( )
```

5.40 src/utls/grading-utils.cpp File Reference

```
#include "grading-utils.h"  
#include <algorithm>  
#include <cstdlib>  
Include dependency graph for grading-utils.cpp:
```



Functions

- int [randomGrade](#) (int min, int max)
- double [calculateFinalGradeMedian](#) (const vector< int > &homeworkGrades, int examGrade)
- double [calculateFinalGradeAverage](#) (const vector< int > &homeworkGrades, int examGrade)

5.40.1 Function Documentation

5.40.1.1 calculateFinalGradeAverage()

```
double calculateFinalGradeAverage (
    const vector< int > & homeworkGrades,
    int examGrade )
```

5.40.1.2 calculateFinalGradeMedian()

```
double calculateFinalGradeMedian (
    const vector< int > & homeworkGrades,
    int examGrade )
```

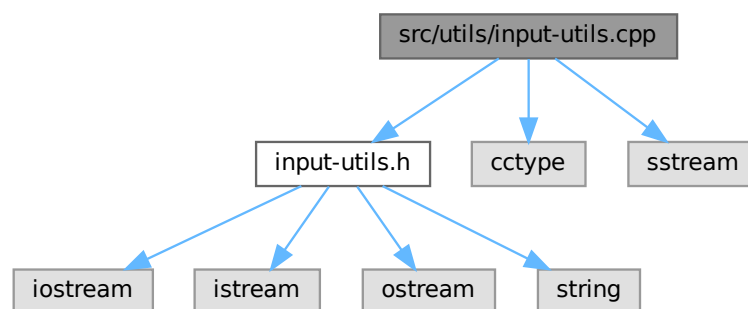
5.40.1.3 randomGrade()

```
int randomGrade (
    int min,
    int max )
```

5.41 src/utls/input-utls.cpp File Reference

```
#include "input-utls.h"
#include <cctype>
#include <sstream>
```

Include dependency graph for input-utls.cpp:



Functions

- int [readIntInRange](#) (const string &prompt, int min, int max)
- string [readSingleStringToken](#) (const string &prompt)
- char [readYesOrNo](#) (const string &prompt)
- int [readSingleIntToken](#) (const string &prompt)

5.41.1 Function Documentation

5.41.1.1 readIntInRange()

```
int readIntInRange (
    const string & prompt,
    int min,
    int max )
```

5.41.1.2 readSingleIntToken()

```
int readSingleIntToken (
    const string & prompt )
```

5.41.1.3 readSingleStringToken()

```
string readSingleStringToken (
    const string & prompt )
```

5.41.1.4 readYesOrNo()

```
char readYesOrNo (
    const string & prompt )
```


Index

- ~Human
 - Human, [8](#)
- ~Student
 - Student, [12](#)
- calculateFinalGradeAverage
 - grading-utils.cpp, [50](#)
 - grading-utils.h, [19](#)
- calculateFinalGradeMedian
 - grading-utils.cpp, [50](#)
 - grading-utils.h, [19](#)
- chooseOutputAndPrint
 - output-results.cpp, [43](#), [44](#)
 - output-results.h, [24](#)
- create-student.cpp
 - createStudentFullyRandom, [33](#)
 - createStudentManual, [33](#)
 - createStudentRandomGrades, [34](#)
 - createStudentsFromFile, [34](#)
 - randomName, [34](#)
 - randomSurname, [34](#)
- create-student.h
 - createStudentFullyRandom, [16](#)
 - createStudentManual, [16](#)
 - createStudentRandomGrades, [16](#)
 - createStudentsFromFile, [16](#)
 - randomName, [16](#)
 - randomSurname, [16](#)
- createStudentFullyRandom
 - create-student.cpp, [33](#)
 - create-student.h, [16](#)
- createStudentManual
 - create-student.cpp, [33](#)
 - create-student.h, [16](#)
- createStudentRandomGrades
 - create-student.cpp, [34](#)
 - create-student.h, [16](#)
- createStudentsFromFile
 - create-student.cpp, [34](#)
 - create-student.h, [16](#)
- examGrade
 - Student, [12](#)
- finalGradeAverage
 - Student, [12](#)
- finalGradeMedian
 - Student, [12](#)
- generate-student-list-test.cpp
 - handleGenerateStudentListTest, [46](#)
- generate-student-list-test.h
 - handleGenerateStudentListTest, [17](#)
- generate-student-list.cpp
 - generateStudentListFile, [40](#)
 - handleGenerateStudentListFile, [40](#)
 - printHeader, [40](#)
 - printStudentListRow, [40](#)
- generate-student-list.h
 - generateStudentListFile, [18](#)
 - handleGenerateStudentListFile, [18](#)
- generateStudentListFile
 - generate-student-list.cpp, [40](#)
 - generate-student-list.h, [18](#)
- grading-utils.cpp
 - calculateFinalGradeAverage, [50](#)
 - calculateFinalGradeMedian, [50](#)
 - randomGrade, [50](#)
- grading-utils.h
 - calculateFinalGradeAverage, [19](#)
 - calculateFinalGradeMedian, [19](#)
 - randomGrade, [20](#)
- handleGenerateStudentListFile
 - generate-student-list.cpp, [40](#)
 - generate-student-list.h, [18](#)
- handleGenerateStudentListTest
 - generate-student-list-test.cpp, [46](#)
 - generate-student-list-test.h, [17](#)
- handleMenuChoice
 - main-menu.cpp, [41](#)
 - student-menu.h, [29](#)
- handleMenuChoice< std::deque< Student > >
 - main-menu.cpp, [41](#)
- handleMenuChoice< std::list< Student > >
 - main-menu.cpp, [41](#)
- handleMenuChoice< vector< Student > >
 - main-menu.cpp, [41](#)
- handleSortNuskriaustiAndProtingiFromFile
 - sort-nuskriausti-and-protingi.cpp, [35](#)
 - sort-nuskriausti-and-protingi.h, [26](#)
- handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > >
 - sort-nuskriausti-and-protingi.cpp, [35](#)
- handleSortNuskriaustiAndProtingiFromFile< std::list< Student > >
 - sort-nuskriausti-and-protingi.cpp, [35](#)
- handleSortNuskriaustiAndProtingiFromFile< vector< Student > >
 - sort-nuskriausti-and-protingi.cpp, [35](#)

handleStudentRuleOfFiveAndOperatorsTest
 student-rule-of-five-operators-test.cpp, 48
 student-rule-of-five-operators-test.h, 30
 homeworkGrades
 Student, 12
 Human, 7
 ~Human, 8
 Human, 8
 name, 9
 name_, 9
 operator=, 9
 setName, 9
 setSurname, 9
 surname, 9
 surname_, 9

 include/create-student.h, 15, 16
 include/generate-student-list-test.h, 17
 include/generate-student-list.h, 18, 19
 include/grading-utils.h, 19, 20
 include/human.h, 20, 21
 include/input-utils.h, 21, 22
 include/output-nuskriausti-and-protingi-test.h, 23
 include/output-results.h, 23, 25
 include/sort-nuskriausti-and-protingi.h, 25, 26
 include/sort-students.h, 27, 28
 include/student-app.h, 28
 include/student-menu.h, 29, 30
 include/student-rule-of-five-operators-test.h, 30, 31
 include/student.h, 31, 32
 input-utils.cpp
 readIntInRange, 51
 readSingleIntToken, 51
 readSingleStringToken, 51
 readYesOrNo, 51
 input-utils.h
 readIntInRange, 22
 readSingleIntToken, 22
 readSingleStringToken, 22
 readYesOrNo, 22

 main
 main.cpp, 43
 main-menu.cpp
 handleMenuChoice, 41
 handleMenuChoice< std::deque< Student > >, 41
 handleMenuChoice< std::list< Student > >, 41
 handleMenuChoice< vector< Student > >, 41
 showMainMenu, 41
 startMainMenu, 41
 startMainMenu< std::deque< Student > >, 42
 startMainMenu< std::list< Student > >, 42
 startMainMenu< vector< Student > >, 42
 main.cpp
 main, 43

 name
 Human, 9
 Student, 12
 name_
 Human, 9

 operator<<
 Student, 14
 student.cpp, 45
 operator>>
 Student, 14
 student.cpp, 45
 operator=
 Human, 9
 Student, 13
 output-nuskriausti-and-protingi-test.cpp
 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles, 47
 outputNuskriaustiAndProtingiTestForContainer, 47
 output-nuskriausti-and-protingi-test.h
 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles, 23
 output-results.cpp
 chooseOutputAndPrint, 43, 44
 outputResults, 44
 output-results.h
 chooseOutputAndPrint, 24
 outputResults, 24
 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles
 output-nuskriausti-and-protingi-test.cpp, 47
 output-nuskriausti-and-protingi-test.h, 23
 outputNuskriaustiAndProtingiTestForContainer
 output-nuskriausti-and-protingi-test.cpp, 47
 outputResults
 output-results.cpp, 44
 output-results.h, 24

 printHeader
 generate-student-list.cpp, 40
 printStudentListRow
 generate-student-list.cpp, 40
 printTestResult
 student-rule-of-five-operators-test.cpp, 48

 randomGrade
 grading-utils.cpp, 50
 grading-utils.h, 20
 randomName
 create-student.cpp, 34
 create-student.h, 16
 randomSurname
 create-student.cpp, 34
 create-student.h, 16
 readIntInRange
 input-utils.cpp, 51
 input-utils.h, 22
 readSingleIntToken
 input-utils.cpp, 51
 input-utils.h, 22
 readSingleStringToken
 input-utils.cpp, 51

- input-utils.h, 22
- readStudent
 - Student, 13
- readYesOrNo
 - input-utils.cpp, 51
 - input-utils.h, 22
- runSortStudentsChoicePrompt
 - sort-students.cpp, 38
 - sort-students.h, 27
- runStudentApp
 - student-app.h, 28
- setExamGrade
 - Student, 13
- setHomeworkGrades
 - Student, 13
- setName
 - Human, 9
 - Student, 13
- setSurname
 - Human, 9
 - Student, 13
- showMainMenu
 - main-menu.cpp, 41
 - student-menu.h, 29
- sort-nuskriausti-and-protingi.cpp
 - handleSortNuskriaustiAndProtingiFromFile, 35
 - handleSortNuskriaustiAndProtingiFromFile<
 - std::deque< Student > >, 35
 - handleSortNuskriaustiAndProtingiFromFile<
 - std::list< Student > >, 35
 - handleSortNuskriaustiAndProtingiFromFile<
 - vector< Student > >, 35
 - sortNuskriaustiAndProtingiFirstStrategy, 35, 36
 - sortNuskriaustiAndProtingiSecondStrategy, 36
 - sortNuskriaustiAndProtingiThirdStrategy, 36, 37
- sort-nuskriausti-and-protingi.h
 - handleSortNuskriaustiAndProtingiFromFile, 26
 - sortNuskriaustiAndProtingiFirstStrategy, 26
 - sortNuskriaustiAndProtingiSecondStrategy, 26
 - sortNuskriaustiAndProtingiThirdStrategy, 26
- sort-students.cpp
 - runSortStudentsChoicePrompt, 38
 - sortStudents, 38, 39
- sort-students.h
 - runSortStudentsChoicePrompt, 27
 - sortStudents, 27
- sortNuskriaustiAndProtingiFirstStrategy
 - sort-nuskriausti-and-protingi.cpp, 35, 36
 - sort-nuskriausti-and-protingi.h, 26
- sortNuskriaustiAndProtingiSecondStrategy
 - sort-nuskriausti-and-protingi.cpp, 36
 - sort-nuskriausti-and-protingi.h, 26
- sortNuskriaustiAndProtingiThirdStrategy
 - sort-nuskriausti-and-protingi.cpp, 36, 37
 - sort-nuskriausti-and-protingi.h, 26
- sortStudents
 - sort-students.cpp, 38, 39
 - sort-students.h, 27
- src/create-student.cpp, 33
- src/data-sorting/sort-nuskriausti-and-protingi.cpp, 34
- src/data-sorting/sort-students.cpp, 37
- src/generate-student-list.cpp, 39
- src/main-menu.cpp, 40
- src/main.cpp, 42
- src/output-results.cpp, 43
- src/student.cpp, 45
- src/test/generate-student-list-test.cpp, 45
- src/test/output-nuskriausti-and-protingi-test.cpp, 46
- src/test/student-rule-of-five-operators-test.cpp, 47
- src/utills/grading-utills.cpp, 49
- src/utills/input-utils.cpp, 50
- startMainMenu
 - main-menu.cpp, 41
 - student-menu.h, 30
- startMainMenu< std::deque< Student > >
 - main-menu.cpp, 42
- startMainMenu< std::list< Student > >
 - main-menu.cpp, 42
- startMainMenu< vector< Student > >
 - main-menu.cpp, 42
- Student, 10
 - ~Student, 12
 - examGrade, 12
 - finalGradeAverage, 12
 - finalGradeMedian, 12
 - homeworkGrades, 12
 - name, 12
 - operator<<, 14
 - operator>>, 14
 - operator=, 13
 - readStudent, 13
 - setExamGrade, 13
 - setHomeworkGrades, 13
 - setName, 13
 - setSurname, 13
 - Student, 11, 12
 - surname, 13
- student-app.h
 - runStudentApp, 28
- student-menu.h
 - handleMenuChoice, 29
 - showMainMenu, 29
 - startMainMenu, 30
- student-rule-of-five-operators-test.cpp
 - handleStudentRuleOfFiveAndOperatorsTest, 48
 - printTestResult, 48
 - testCopyAssignment, 48
 - testCopyConstructor, 48
 - testDestructor, 48
 - testHumanIsAbstractAndNotInstantiable, 48
 - testInputOperatorInvalid, 48
 - testInputOperatorValid, 48
 - testMoveAssignment, 49
 - testMoveConstructor, 49
 - testOutputOperator, 49
- student-rule-of-five-operators-test.h

- handleStudentRuleOfFiveAndOperatorsTest, [30](#)
- student.cpp
 - operator<<, [45](#)
 - operator>>, [45](#)
- surname
 - Human, [9](#)
 - Student, [13](#)
- surname_
 - Human, [9](#)
- testCopyAssignment
 - student-rule-of-five-operators-test.cpp, [48](#)
- testCopyConstructor
 - student-rule-of-five-operators-test.cpp, [48](#)
- testDestructor
 - student-rule-of-five-operators-test.cpp, [48](#)
- testHumanIsAbstractAndNotInstantiable
 - student-rule-of-five-operators-test.cpp, [48](#)
- testInputOperatorInvalid
 - student-rule-of-five-operators-test.cpp, [48](#)
- testInputOperatorValid
 - student-rule-of-five-operators-test.cpp, [48](#)
- testMoveAssignment
 - student-rule-of-five-operators-test.cpp, [49](#)
- testMoveConstructor
 - student-rule-of-five-operators-test.cpp, [49](#)
- testOutputOperator
 - student-rule-of-five-operators-test.cpp, [49](#)